

José R. Sousa

Charlotte, NC 28217

LinkedIn: <https://www.linkedin.com/in/jrsousa2/>

Phone: 904-290-2890

E-mail: jrsousa2@gmail.com

GitHub: <https://jrsousa2.github.io/>

SUMMARY OF QUALIFICATIONS

- Large experience in **database / data warehouse** and **SQL** (query language).
- Great with **Python** programming in **VS Code**. Created executable from code, developed **REST API's**, created **GUI's**, , loaded **Excel** and **XML** files into **pandas** dataframes, read and wrote to Excel files with **openpyxl**, created formatted **PowerPoint** files automatically, created **virtual environments**, automated **batch jobs**, etc. **All can be viewed on my GitHub**.
- Extracted data from **Hadoop** with Dremio / Hive and created VDS's.
- Fitted **logistics regression** model for delinquency risk using **PySpark ML**.
- Created **AVM** to predict bare land value of properties in Orange county, CA.
- Independently created **logistics regression** model to score risk of auto policyholders. The model was deployed.
- Worked on **insurance pricing models** in partnership with consultancies.
- Created **workflows** in **n8n** for Gen AI.
- Exposure to Deep Learning and NLP modules in Python.
- **Expert at SAS** (**Enterprise Guide** and **Viya**) -- created **stored processes** and **ETL's**, wrote advanced code to produce tables and reports, and truly mastered **SAS macros**. Created dashboard with **GRAPH** and reports with **Proc Report**.
- Strong with **Machine Learning** models and **data analytics** and familiar with **actuarial analyses** and **actuarial reports**.
- Expert at **MS Excel**, created advanced spreadsheets with **VBA macros**, **automated charts and graphs**.
- Most of my experience is in the **insurance** industry (10+ years). Created a wide variety of **loss triangles**, estimated **ultimate claims** from LDF's, calculated ceded losses from **reinsurance** contracts, calculated earned premiums, created reports on **loss ratio** and **in-force policies**, built **predictive models**, etc.
- Exposure to **accounting principles** (GAAP, STAT), created **YTD summary reports** to support quarterly filings (10-Q and 10-K) using the **general ledger** database and performed account and profit center level reconciliations with it.
- Graduated in **Math / Statistics** from the **best university in Brazil** (**U of São Paulo**).
- Very analytical, self-taught, can **work independently**, created new formulas in math (e.g. <https://arxiv.org/html/1810.07877v7>).
- **U.S. citizen** (had a **green card** through **NatGen** previously).

SOFTWARE

- **Databases:** **Oracle**, **Teradata**, **SQL Server / SSMS** and their **IDE's**.
- **Big Data:** **Snowflake**, **Databricks**, **PySpark**, **Hadoop**, **Dremio**, **Hive**.
- **Programming:** **Python**, **VBA**, **VBScript**, **Unix/Windows batch scripting**.
- **Cloud:** Working knowledge of **AWS**, **Azure** and **GCP**.
- **SAS:**
 - Platforms: **Unix**, **Linux**, **Windows Server** and **Viya** (Cloud).
 - Modules: **BASE**, **STAT**, **GRAPH**, **ACCESS**, **CONNECT**.
 - Expert knowledge of most used SAS Procs, including **Proc Report**. Advanced **macro language** and **stored processes**.
- **BI / Predictive Modeling:** **SAS Enterprise Guide**, **Power BI**.
- **Code Editors:** **VS Code**, **Jupyter Notebook**.
- **Other:** **Excel**, **PowerPoint**, **SharePoint**, **Git**, **RegEx**, **Jira**, **XML**, **JSON**.

PROFESSIONAL EXPERIENCE

Wells Fargo (**banking**)

▪ Second project: 2025

- Worked on AML / Financial Crimes remediations (CIP / BO / CDD / KYC).
- Built AML remediation populations (Trust, NRM4 and transportation assets).
- Created population identification workbooks and waterfall charts and presented them to senior directors for approval.
- Wrote complex optimized SQL queries to extract data from Hadoop using Dremio and Hive and from various databases (Teradata, Oracle, SQL Server, DB2, etc.). Was a go-to person for challenging technical questions.
- Developed SAS and SQL code to create ad-hoc, time-sensitive data extracts and reports.
- Created code test scripts and worked with code testers to validate the code.
- Recognized with a full-time offer at project completion.

▪ First project: 2022 through 2025

- Worked on customer remediations, responsible for the creation of the final artifacts from which refunds and letters are issued to customers as required by regulators, following strict Confluence procedures.
- Managed SAS code used to create final remediation artifacts, made changes to the code to add enhancements, optimize performance and minimize the number of defects and findings in peer reviews. As an example of enhancement, when a new analyst takes over a remediation, all the artifacts and SAS tables created by the previous analyst are automatically saved to a new subfolder as backup.
- Created ETL's to produce new artifacts by mapping fields from the execution data requirements (EDR) to data tables. Contributed to design artifact layouts.

Truist, Charlotte, NC (banking)

Data Analytics SQL/Python (contract / remote)

09/2021 – 03/2022

- Created rules based decision tool for the compliance and data analysts to choose their optimal analytics application as a web form created in Python.
- Created and maintained current report on user accesses for the various proprietary applications and data repositories (AD, ACL), for compliance.
- Ran reconciliation between ServiceNow/RITM and data pulled from the mainframe to assess correctness of an automatic ticket creation process for abends (abnormal job termination), and presented findings with PowerPoint.
- Ran SQL queries in Snowflake to extract data from the data lake/data warehouse.
- Created reports using the SAS grid and produced documentation.

Florida Blue, Jacksonville, FL (healthcare)

SAS/Python Developer (contract / remote)

04/2020 – 08/2021

- Developed ETL code to extract data from Oracle and DB2 and others, using SAS 9.4 on AIX (Unix) and Enterprise Guide.
- Extracted data from multiple sources to create agent fraud detection analytics for the SIU dept with various metrics. Created macro to automate the formatting of the report for each metric using Proc Report.
- Automated and scheduled various SAS ETL jobs in Unix, using a SAS macro.
- Used HTML syntax in SAS to compose rich text e-mail alerts.
- Combined Windows batch, WinScp and VB scripts to schedule jobs and automate file transfers and moving data between platforms using SFTP.
- Created a neat SAS stored process for various canned reports.
- Created visualizations and reports using Power BI and Python.

- Gave **training** to team members on how to design **SAS stored processes**, covering how to create the **GUI** and how to handle the variable **user's inputs**.

National General (Allstate), New York, NY (insurance)

Insurance Data Programmer / Data Manager

04/2015 – 12/2019

- **Reported to the holding company's chief actuary.**
- Mastered the **claims data system** of the company (EPIC) and legacy data stored in **Oracle** and **SQL Server** databases.
- **Planned** and **created** datasets in **SAS** by extracting and combining data from various databases, from which loss triangles and **data analytics** were produced.
- Created **actuarial loss triangles** used in reserving analyses with **SAS macros**, and reconciled the numbers with the company's **general ledger**.
- Created quarterly company-wide **YTD summary report** in **Excel** using **general ledger data** extracted with **Oracle BI Discoverer** to support quarterly filings and the annual report preparation (**10K**, **10Q**), for the **SEC** and other **regulators**.
- Participated in the **ATLAS project**, with a goal to capture claim and policy level data into the **GL Oracle database**.
- Extracted data from the **SAS server** to create reports with **SAS macros** and **code**. These reports were used internally or provided to consultants, **auditors** and **regulators**.
- Using **SAS**, ran **statistical data analyses** to track performance of products.
- Created complete documentation about work done on legacy data from acquired companies.

AIG, New York, NY (insurance)

Data Scientist / SAS Programmer (contractor / FTE)

12/2011 – 12/2014

- Planned and extracted data from multiple sources (**mainframe**, **DB2**, **SQL Server** and **Oracle**) to produce **enhanced tables** used in all **reports** and **models** and **processes**, using **ETL** and **SAS/Access** interface to **OLE DB** and **ODBC**.
- Created **ETL's** to read **VB/VSAM** files on **IBM mainframe** and load them into **SAS** tables on Windows. These files became the data foundation of the **Global Actuarial Pricing** department.
- Wrote **SAS macro** to create a wide variety of **loss triangles** used to derive **LDF's** and **estimate** the **ultimate claims**.
- Worked in the **deep dive** and **profitability analyses** of the book of business, by extracting data on **losses**, **premium**, **policy** and **insured attributes** to generate various reports and metrics and to monitor the **performance** of the various **lines of business and segments**.
- Created smart **algorithm for linking policies** based on insured and policy characteristics in **SAS**. This policy linking was the basis for the creation of a **scoring model** called **AQI (Account Quality Index)**.
- Worked with Deloitte consultants and provided them with **loss triangles** used as inputs into the Commercial Auto **pricing tool**. Also worked with Deloitte on the **Specialty Worker's Comp model**, making sure that deliverables such as **SAS code**, **data** and results of the **predictive model** met the requirements.
- Created and scheduled weekly **jobs** to extract submission data entered by underwriters into the **AQI model** and update a **Qlikview dashboard**.

EDUCATION

Masters's in **Statistics** - **University of São Paulo, Brazil**

Bachelor's in **Statistics** - **University of São Paulo, Brazil**

Bachelor's in **Mathematics** - **University of São Paulo, Brazil**